

# Git & GitHub Ultimate Cheat Sheet (2026 Edition)

A one-page (expanded) cheat sheet covering the Git commands you use 95% of the time. Perfect for students, interviews, and daily development.

## 1. Configure Git (First Time Only)

```
git config --global user.name "Your Name"
git config --global user.email "you@example.com"

git config --global init.defaultBranch main

git config --global core.editor "code --wait"

git config --global color.ui auto

git config --list
```

## 2. Create Repository

Initialize new repository

```
git init
```

Clone existing repository

```
git clone https://github.com/user/project.git
```

Clone into another folder

```
git clone https://github.com/user/project.git myproject
```

Clone a specific branch

```
git clone -b development https://github.com/user/project.git
```

## 3. Repository Information

Current status

```
git status
```

### Repository history

```
git log
```

### Compact history

```
git log --oneline
```

### Beautiful graph

```
git log --oneline --graph --all
```

### Last commit

```
git log -1
```

## 4. Add Files

### Add one file

```
git add app.py
```

### Add multiple files

```
git add file1.py file2.py
```

### Add entire folder

```
git add folder/
```

### Add everything

```
git add .
```

### Interactive add

```
git add -p
```

## 5. Commit

### Commit

```
git commit -m "Initial commit"
```

### Add + Commit

```
git commit -am "Updated feature"
```

### Modify previous commit

```
git commit --amend
```

## 6. Branch Commands

### Create branch

```
git branch feature-login
```

### List branches

```
git branch
```

### List remote branches

```
git branch -r
```

### List all branches

```
git branch -a
```

### Delete branch

```
git branch -d feature-login
```

### Force delete

```
git branch -D feature-login
```

### Rename branch

```
git branch -m old new
```

## 7. Switch Branches

```
git switch main
```

```
git switch feature-login
```

```
git checkout feature-login
```

Create and switch

```
git switch -c feature
```

Old syntax

```
git checkout -b feature
```

## 8. Merge

Merge branch

```
git merge feature-login
```

Fast-forward merge

```
git merge --ff-only feature-login
```

Abort merge

```
git merge --abort
```

## 9. Rebase

Rebase

```
git rebase main
```

Interactive rebase

```
git rebase -i HEAD~5
```

Continue

```
git rebase --continue
```

Abort

```
git rebase --abort
```

Skip

```
git rebase --skip
```

.

## 10. Remote Repository

Show remotes

```
git remote -v
```

Add remote

```
git remote add origin URL
```

Change URL

```
git remote set-url origin URL
```

Remove remote

```
git remote remove origin
```

.

## 11. Push

First push

```
git push -u origin main
```

Normal push

```
git push
```

Push branch

```
git push origin feature
```

Delete remote branch

```
git push origin --delete feature
```

Force push

```
git push --force
```

Safe force push

```
git push --force-with-lease
```

## 12. Pull

Pull latest

```
git pull
```

Pull specific branch

```
git pull origin main
```

Pull using rebase

```
git pull --rebase
```

## 13. Fetch

```
git fetch
```

```
git fetch origin
```

```
git fetch --all
```

## 14. Compare

Compare files

```
git diff
```

Staged files

```
git diff --staged
```

Compare branches

```
git diff main feature
```

Compare commits

```
git diff COMMIT1 COMMIT2
```

# 15. Restore

Restore file

```
git restore app.py
```

Restore staged

```
git restore --staged app.py
```

Restore everything

```
git restore .
```

# 16. Reset

Soft reset

```
git reset --soft HEAD~1
```

Mixed reset

```
git reset HEAD~1
```

Hard reset

```
git reset --hard HEAD~1
```

Reset specific file

```
git reset app.py
```

# 17. Stash

Save changes

```
git stash
```

Named stash

```
git stash push -m "work"
```

List stashes

```
git stash list
```

### Apply

```
git stash apply
```

### Pop

```
git stash pop
```

### Delete stash

```
git stash drop
```

### Delete all

```
git stash clear
```

## 18. Tags

### Create tag

```
git tag v1.0
```

### Annotated tag

```
git tag -a v1.0 -m "Release"
```

### Push tag

```
git push origin v1.0
```

### Push all tags

```
git push origin --tags
```

### Delete tag

```
git tag -d v1.0
```

## 19. Cherry Pick

```
git cherry-pick COMMIT_ID
```



Abort

```
git cherry-pick --abort
```

Continue

```
git cherry-pick --continue
```

## 20. Reflog

Show history

```
git reflog
```

Recover

```
git reset --hard HEAD@{3}
```

## 21. Clean Repository

Preview

```
git clean -n
```

Delete

```
git clean -f
```

Delete folders

```
git clean -fd
```

## 22. Ignore Files

Create

```
.gitignore
```

Example

```
__pycache__/  
venv/  
.env
```

```
node_modules/
```

```
dist/
```

```
build/
```

```
*.log
```

```
*.pyc
```

```
.idea/
```

```
.vscode/
```

## 23. Show Information

Who changed what

```
git blame file.py
```

Show commit

```
git show COMMIT
```

Show branch

```
git branch
```

Current commit

```
git rev-parse HEAD
```

## 24. Undo Mistakes

Undo last commit

```
git reset --soft HEAD~1
```

Undo pushed commit

```
git revert COMMIT_ID
```

Undo staging

```
git restore --staged file.py
```

Undo modifications

```
git restore file.py
```

## 25. GitHub Commands

Clone

```
git clone URL
```

Fork

(Use GitHub UI)

Create Pull Request

(GitHub UI)

Merge Pull Request

(GitHub UI)

Review PR

(GitHub UI)

Approve PR

(GitHub UI)

## 26. GitHub Flow

Clone

↓

Create Branch

↓

Code

↓

git add  
↓  
git commit  
↓  
git push  
↓  
Pull Request  
↓  
Code Review  
↓  
Merge  
↓  
Delete Branch  
,

## 27. Git Flow

main  
↓  
develop  
↓  
feature branch  
↓  
develop  
↓  
release  
↓  
main  
↓  
hotfix

↓

main

## 28. Most Used Commands (Top 20)

```
git init
git clone URL
git status
git add .
git commit -m "message"
git log --oneline
git branch
git switch branch
git switch -c newbranch
git checkout
git merge
git rebase
git stash
git stash pop
git fetch
git pull
git push
git remote -v
git diff
git restore
```

## 29. Complete Daily Workflow

```
git clone URL
git switch -c feature-login
```

```
# Write code

git status

git add .

git commit -m "Added Login API"

git push -u origin feature-login

# Create Pull Request

# After merge

git switch main

git pull

git branch -d feature-login
```

## 30. Emergency Recovery Commands

```
git reflog

git reset --hard HEAD@{1}

git merge --abort

git rebase --abort

git cherry-pick --abort

git stash

git stash pop

git restore .

git restore --staged .

git clean -fd
```

## Git Command Categories (Quick Reference)

| Category   | Commands            |
|------------|---------------------|
| Repository | git init, git clone |

| Category          | Commands                                  |
|-------------------|-------------------------------------------|
| Configuration     | git config                                |
| Status            | git status, git log                       |
| Staging           | git add                                   |
| Commit            | git commit, git commit --amend            |
| Branching         | git branch, git switch, git checkout      |
| Merging           | git merge                                 |
| Rebasing          | git rebase                                |
| Remote            | git remote, git fetch, git pull, git push |
| Compare           | git diff                                  |
| Undo              | git restore, git reset, git revert        |
| Temporary Save    | git stash                                 |
| Releases          | git tag                                   |
| Selective Commits | git cherry-pick                           |
| Recovery          | git reflog                                |
| Cleanup           | git clean                                 |
| Investigation     | git blame, git show                       |